GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 13



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Time & Distance

Topics to be Covered



Topic-1 Problem on Trains

Topic-2 Boats and Streams

Topic-3 Races

[MCQ]

$T = \frac{D}{S}$

#Q. A man misses a train by 40 minutes if he travels at 30 kmph. If he travels at 40 kmph, then also he misses the train by 10minutes. What is the minimum speed required to catch the train on time?

A.T. $= 1\text{hrs } 20\text{mins} = 1\frac{20}{60} = 1\frac{1}{3} = \frac{4}{3}\text{hrs}$

$\frac{x}{30} - \frac{x}{40} = \frac{1}{2}$

$\Rightarrow \frac{x}{120} = \frac{1}{2} \quad \therefore x = 60\text{km}$

Assignment

Time $= \frac{60}{30} = 2\text{hrs}$

$S = \frac{15}{60} \times 3 = \frac{1}{4} = 45\text{km/hr}$

A 50 km/hr

B 55 km/hr

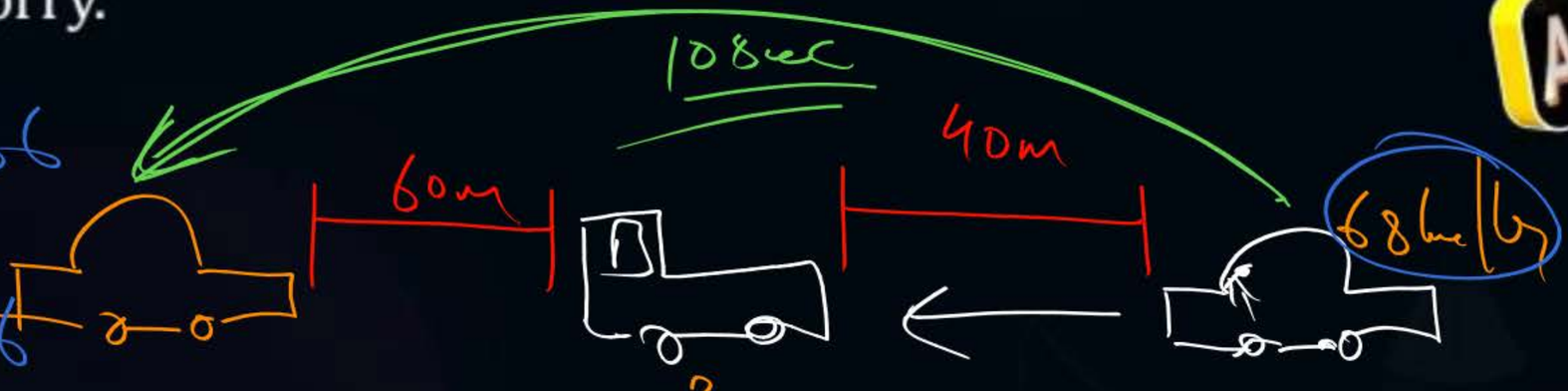
C 60 km/hr

D 45 km/hr

[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

Assignment



S.C − S.L = 36

68 − S.L = 36

S.L = 68 − 36

= 32 km/hr

D = 100m

T = 10second

Relative speed = $\frac{100}{10}$ = 10m/sec × $\frac{18}{5}$ = 36 km/hr

[MCQ]

#Q. Two boys begin together to write out a booklet containing 817 lines. The first boy starts with first line, writing at the rate of 200 lines an hour and the second boy starts with last line. He writes line 817 and so on backwards proceeding at the rate of 150 lines an hour. At what line will they meet?

A. 469th

B. 467th

C. 465th

D. 463rd

$D = 817$ lines

$R.S. = 350$ lines/hr

$\text{Time} = \frac{817 \text{ lines}}{350}$ hr

Assignment

$D = \frac{4}{200} \times \frac{817}{350}$

$= \frac{3268}{7} = 466.85\overline{7}$

467th

Meeting at a point:

$D_1 + D_2 = T.D.$

$D_1 = D_2$

Two objects moving in opposite directions towards each other:
Distance covered by 1st object + Distance covered by 2nd object = Total distance





Two objects moving in same directions :
Distance covered by 1st object = Distance covered by 2nd object





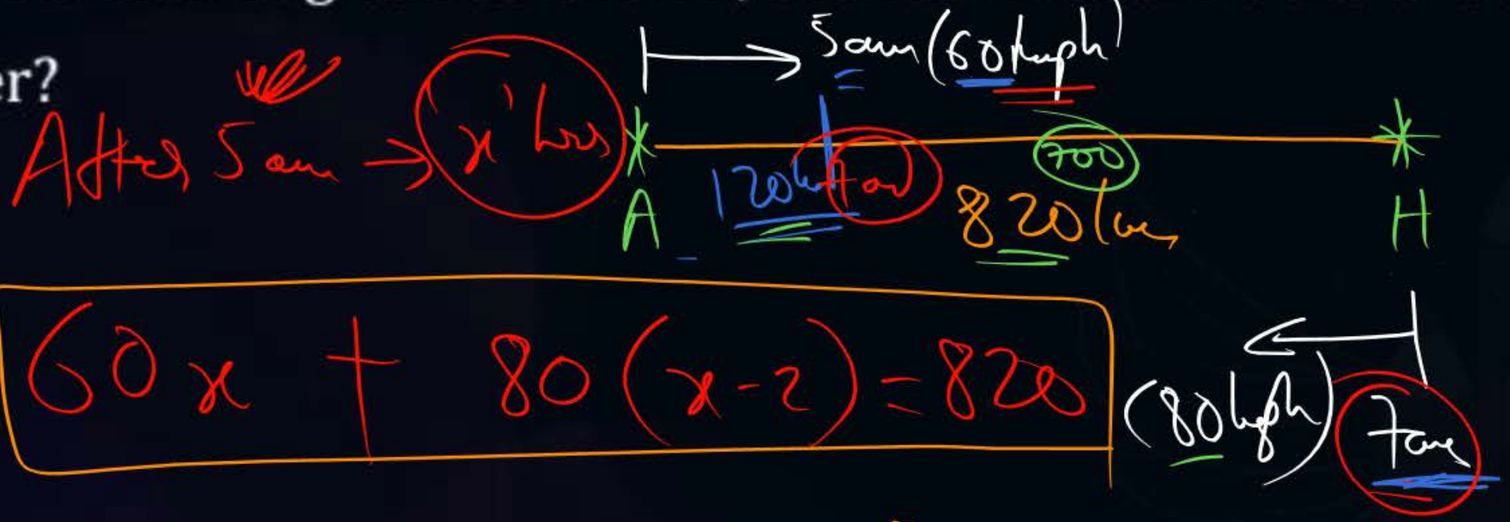
[MCQ]

5am + 7hrs = 12PM

$D_1 + D_2 = T \cdot D$

7am + 5hrs = 12PM

#Q. A train starts from Ahmedabad to Howrah at 5am everyday with the speed of 60kmph. Another train starts from Howrah to Ahmedabad at 7am everyday with the speed of 80 kmph. If the distance between Ahmedabad and Howrah is given as 820km, then at what time the two trains meet each other?



$60x + 80(x-2) = 820$

$\Rightarrow 60x + 80x - 160 = 820$

$\Rightarrow 140x = 980$

$\therefore x = 7\text{hrs}$

Assignment

$D = 700\text{km}$

$R.S. = 140\text{km/h}$

$T = \frac{700}{140} = 5\text{hrs}$

[MCQ] 11+9=20

$D_1 = D_2$

#Q. A train starts from Chennai to Delhi at 11am with the speed of 110kmph. Another train starts from Chennai to Delhi at 9am the same day with the speed of 90 kmph. If the distance between Chennai and Delhi is given as 1920km, then at what time the two trains meet each other?

8PM

After 9AM $\Rightarrow$ 'x' hrs



$90x = 110(x-2)$

$\Rightarrow 90x = 110x - 220$

$20x = 220$

$x = 11$ hrs

Assignment

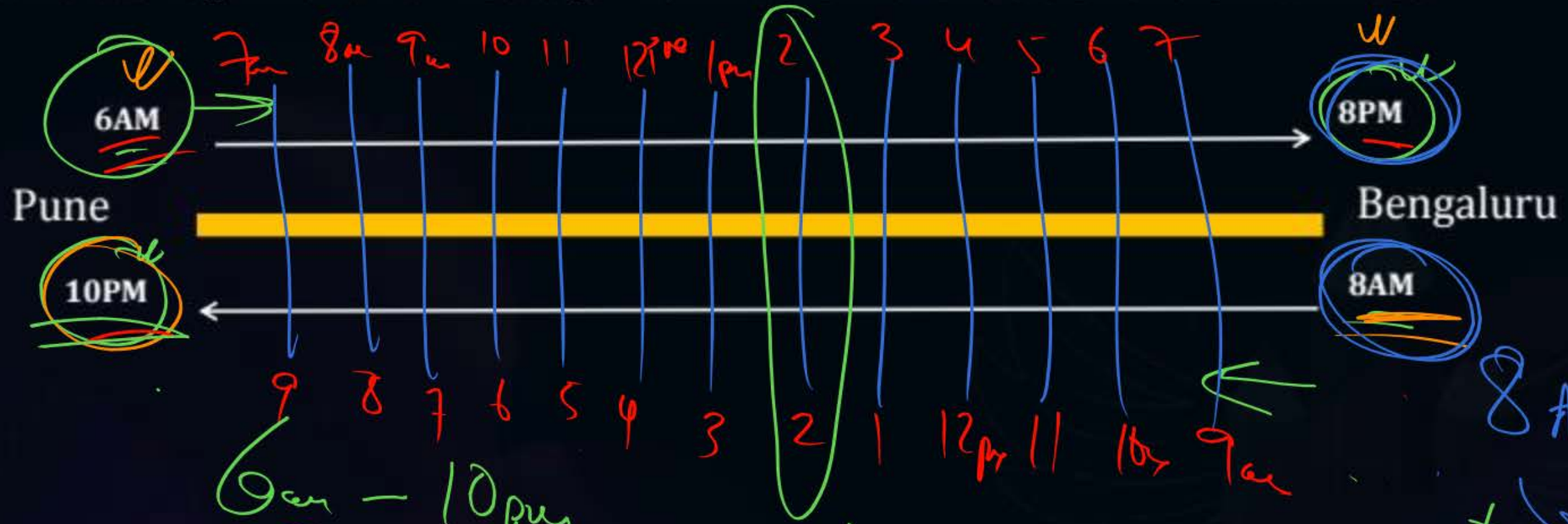
D = 180km

R.S. = 20km/hr

$T = \frac{180}{20} = 9$ hrs

[MCQ]

#Q. A train starts from Pune at 6am everyday and reaches Bengaluru 8pm the same day. Another train starts from Bengaluru at 8am everyday and reaches Pune at 10pm the same day. At what time these two trains meet?



[MCQ]

#Q. A train starts from Leh at 5am everyday and reaches Kanyakumari at 9am the same day. Another train starts from Kanyakumari at 8am everyday and reaches Leh at 2pm the same day. At what time these two trains meet?



$D_1 + D_2 = D$

$\frac{3}{4} \times 60 = 45$ minutes

$S_{T_1} = \frac{x}{4}$

$S_{T_2} = \frac{x}{6}$

After 5 am → 'y' hrs

$\frac{x}{4} \times (y) + \frac{x}{6}(y-3) = x$

$\frac{y}{4} + \frac{y-3}{6} = 1$

$\Rightarrow 3y + 2y - 6 = 12$

$\Rightarrow 5y = 18$

$y = \frac{18}{5} = 3\frac{3}{5}$ hrs

$= 3$ hrs 36 min

8:36 AM

Trains

D = length of train



D = l of Train + l of Platform

Time = $\frac{\text{Dist}}{\text{Speed}}$

$\frac{\text{l of Train}}{\text{Speed}}$

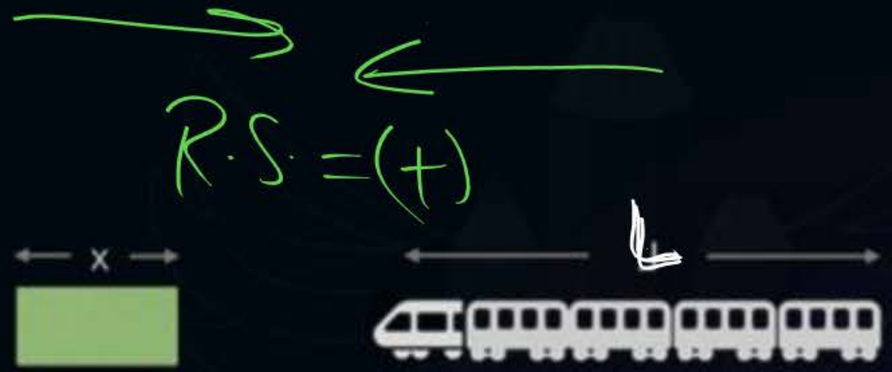


Trains

When a train try to cross a signal pole or standing man completely...
D = length of the train



When a train try to cross a Platform or Bridge or Tunnel.....
D = length of the train + length of Platform or Bridge or Tunnel



Trains

When a train try to cross another moving train.....

SAME DIRECTION : Speed (-)
OPPOSITE DIRECTION : Speed (+)

Distance = Length of Train 1 + Length of Train 2



[MCQ]

#Q. If a train travelling at 40 km/hr crosses another train of length 100 meter travelling at 14 km/hr in opposite directions in 30 seconds, then find the length of the train. (meters)

$l_{T_1} + 100 = 15 \times 30$

$\Rightarrow l_{T_1} = 450 - 100$

$= 350\text{m}$

R.S. $= 40 + 14 = 54\text{ km/hr}$

$\times \frac{5}{18}$

Time $= 30\text{ sec}$ $= 15\text{m/sec}$

[MCQ]

$D = S \times T$

#Q. A train running at 52 km/hr takes 36 seconds to pass a platform. Next it takes 24 seconds to cross a man walking at the platform with 10 km/hr in the same direction. Find the length of the platform. (meters)

Train crossing PLATFORM

$280 + l_P = 52 \times \frac{5}{18} \times 36$

$\Rightarrow 280 + l_P = 520$

$\therefore l_P = 520 - 280 = 240\text{m}$

Train Crossing MAN

$D = l \text{ of Train}$

$R.S. = (42)\text{km/hr} \Rightarrow (42) \times \frac{5}{18}\ \text{m/sec}$

$T = 24\text{sec}$

$l \text{ of Train} = 14 \times \frac{5}{18} \times 24 = 280\text{m}$

Boats & Streams

$$S.B. = \frac{D.S + U.S}{2}$$

$$S.W. = \frac{D.S - U.S}{2}$$

Downstream speed (DS)

Upstream speed (US)



→ 8 km/hr

6 km/hr

10 km/hr

$$S.B + S.W = D.S$$

$$S.B - S.W = U.S$$



→ 8 km/hr

→ 2 km/hr



[MCQ]

#Q. A man rows his boat downstream @ 18 km/hr & upstream @ 10 km/hr.

Find the speed of boat in still water.

Speed of Boat = $\frac{D.S + U.S}{2}$

$= \frac{18 + 10}{2} = \frac{28}{2}$

= 14 km/hr

A 12 km/hr

B 10 km/hr

C 14 km/hr

D 16 km/hr



[MCQ]

#Q. A boat traveled from A to B and back to A from B in 5 hours. If the speed of boat in still water and the speed of stream be 7.5 kmph and 1.5 kmph, then what is the distance between A and B?

Assignment

A 18 km

B 10 km

C 16 km

D 5 km

Races

Assignment

#Q. In a km race A defeats B by 100 meters and B defeats C by 50 meters, then A defeats C by how many meters?





2 mins Summary

Topic Trains, Boats & Races

THANK - YOU